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How Does Cost Leadership Strategy Suppress the Performance Benefits of Explainability of AI Applications in Organizations?

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ABSTRACT

This study examines how explainable artificial intelligence-enabled agility, encompassing operational and market agility, influences performance. We examine the potentially negative effect of a cost-leadership strategy on the performance benefits of explainable AI. Data from a survey of IT executives was analyzed. Findings suggest that the explainability of artificial intelligence applications does not directly relate to performance. Instead, its benefits are actualized through operational and market agility. However, our study finds that in organizations that prioritize cost leadership, the positive link between explainability and performance becomes negative and significant. Additionally, the mediating roles of operational and market agility become insignificant in such organizations.

KEYWORDS

Explainable Artificial Intelligence, Operational Agility, Market Agility, Cost Leadership, Performance

INTRODUCTION

Despite the proven performance benefits of artificial intelligence (AI) applications at both operational and market agility levels (Liu & Li, 2022), the complexity of AI algorithms has led to a perception of AI as a "black box." In response to this challenge, explainable AI (XAI) emerges as a potential solution. Explainability of AI applications is defined as: "Given a certain audience, explainability refers to the details and reasons a model gives to make its functioning clear or easy to understand" (Arrieta et al., 2020, p.82).

AI capabilities can improve agility within the organization as a mediator to improve performance (Dubey et al., 2022; Wamba, 2022; Shafiabady et al., 2023). One overlooked aspect is how the

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Table 1. Terms similar to explainability

Term	Definition	Reference
Comprehensibility	When applied to machine learning models, comprehensibility refers to the capacity of a learning algorithm to present its acquired knowledge in a manner that is understandable to humans.	Fernandez et al., (2019); Gleicher (2016)
Transparency	A model is deemed transparent if it is inherently understandable on its own.	Ehsan et al. (2021); Maarten Schraagen et al. (2021)
Interpretability	The degree to which a human can understand and replicate the rationale behind a decision made by AI.	Linardatos et al. (2021); Miller (2019)
Explainability	"Given a certain audience, explainability refers to the details and reasons a model gives to make its functioning clear or easy to understand."	Arrieta et al. (2020, p.82)

capabilities acquired by firms through the explainability of their AI applications and mediators, such as operational and market agility, lead to superior performance (Hasan et al., 2023; Li et al., 2022). This study also explores the role of costs of XAI applications and the cost leadership strategy (CLS) in this mediated relationship. Cost leadership (CL) is a competitive strategy whereby a company positions itself as the most cost-effective producer or provider of a particular product or service in a market (Chelliah et al., 2023). A range of studies have examined the costs of implementing XAI in organizations (Dessain et al., 2023; Höhn & Faradouris, 2021). We posit that if an organization adopts CL, the performance benefits of XAI are influenced. In this study we aim to investigate how the CLS influences the extent to which the explainability of AI applications generates performance through operational and market agility. This study examines the moderated role of CL in the mediation of agility between the explainability of AI applications and firm performance. We present the following research question: To what extent does an organization's CLS impact on the performance benefits gained from the explainability of its AI applications through agility?

Although existing literature has explored the technological advancements of XAI (Alnsour et al., 2023; Hassija et al., 2024), there is a gap for how the mediating factors of operational and market agility interact with the explainability of AI applications to influence firm performance. This study is the first that investigates the mediation of agility in relation to explainability of AI applications. A stream of literature (Amoako-Gyampah & Acquaah, 2008; Hilman & Kaliappen, 2014; Kaliappen & Hilman, 2013; Kharub et al., 2022; Nandakumar et al., 2010; Voola & O'Cass, 2010) looks at the association between a firm's decision to implement a CLS and a firm's performance. There is also literature on the positive moderating role of CL in enhancing organizational performance (Al-Rdaydeh, 2018; Alyoubi & Yamin, 2021; Yusuf et al., 2020). This study addresses the gap in the literature by investigating the negative moderating role of CL in organizations that must deal with XAI costs. To address these two gaps, our study has been grounded through the lenses of dynamic capability (DC) view and contingency theory (CT).

LITERATURE REVIEW AND THEORETICAL BACKGROUND

Explainability of AI Applications

Multiple definitions exist for explainability of an AI system. In Table 1, we clarify the differences and similarities among four frequently used terms in the field of XAI.

Across these definitions, understandability emerges as a central concept in XAI. Understandability in this context encompasses both the model and human sides. For instance, comprehensibility and transparency relate to a model's ability to be inherently understandable, whereas interpretability

gauges how well a human can comprehend a model's decision. This two-sided perspective on terms used in the field of XAI emphasizes the importance of considering the audience's cognitive skills and goals alongside the model's characteristics. This viewpoint underscores the audience as a key element in XAI.

Building upon the philosophical definitions (Sovrano et al., 2021; Sovrano & Vitali, 2023) presented in Appendix P, we refer to "the details or reasons that someone gives to make something clear or easy to understand," as noted by Cambridge Dictionary, for a definition of explanation. In the context of AI, this can be rephrased as the details or reasons that an AI system offers to elucidate or simplify its results. However, this definition raises two ambiguities. First, the reasons provided for an explanation depend entirely on the audience to whom they are addressed. Second, the effectiveness of the explanation in making the concept clear or easy to understand also hinges entirely on the audience. A modified perspective proposed by Arrieta et al. (2020) defines explainability for AI applications as "given a specific audience, explainability pertains to the specifics and rationale that a model provides to clarify or simplify its operations" (p.82) Following this definition, Arrieta et al. (2020) defined XAI as "Given an audience, an explainable Artificial Intelligence is one that produces details or reasons to make its functioning clear or easy to understand" (p.82).

Costs for Explainability of AI Applications

Maxwell et al. (2020) identified seven categories of XAI cost that are presented in Appendix M along with their cost factors. Höhn and Faradouris (2021) examined XAI costs for organizations and looked at the costs and requirements of explainability for AI applications. Dessain et al. (2023) investigated the expenses associated with incorporating explainability into machine learning models for credit scoring. Rieger et al. (2023) stated that in scenarios where errors are common, explainability can negatively impact human-AI interaction, exposing organizations to additional costs. This statement highlights the dual nature of explainability as both beneficial and potentially costly. Akyol et al. (2016) made the initial effort to quantitatively assess the cost of transparency in explainable ML applications.

Cost Leadership and Performance

When a business establishes itself as the most cost-effective producer or provider of a particular product or service in a market, it is practicing CL. This strategy demands ongoing cost reduction efforts across organizational levels to sustain competitiveness, viewed as a significant advantage (Chelliah et al., 2023). Porter (1985) suggested that adopting the CLS can help a firm establish a "low-cost position," which serves as a defense against competitors. Lower costs enable the firm to maintain profitability even when competitors struggle to do so. Ultimately, being in a low-cost position typically places the firm in a favorable position relative to its competitors in the industry. Extensive literature examined the relationship between CLS and firm performance. Table 2 presents the literature on association between CL and firm performance.

Table 3 summarizes the impact of AI capabilities on effectiveness of CL in improving performance. For example, Ergün and Karabulut (2023) and Wirtz (2020) explored the role of AI as a mediator for the effect of CL on achieving competitive advantage. Similarly, Mithas et al. (2020) highlighted the effect of AI for successful implementation of CL to outperform. Jerab and Mabrouk (2023) demonstrated the role of CL as moderator in association between CL and performance. Although the literature addresses the role of AI in association between CL and performance, in this study we focus on the impact of CL in achieving performance by implementing XAI.

Agility

Agility is defined as an organization's capacity to effectively and efficiently redeploy resources to value-creating activities (Teece et al., 1997). In IT capabilities, agility involves both technical and managerial aspects, fostering flexibility and integration for sensing and responding to a business environment (Batta et al., 2023; Xie et al., 2022). We adopt the definition of agility presented by

Table 2. Summary of selected research

Study	Context of the study	Other constructs
Kharub et al. (2022)	Mediating role of manufacturing strategies in the relationship between competitive strategies and firm performance.	Mediators:
		Cost
		Delivery
		Flexibility
		Quality
Hilman and Kaliappen (2014)	Connection between CLS, process innovation, and organizational performance.	Mediator:
		Process innovation
Kaliappen and Hilman (2013)	Connection between CLS, competitor orientation, and organizational performance.	Mediator
		Competitor orientation
Voola and O'Cass (2010)	The relationship among competitive strategies and firm performance.	Mediators:
		Responsive market orientation
		Proactive market orientation
Nandakumar et al. (2010)	The relationship between business-level strategy and organisational performance.	Moderators:
		External environment
		Organizational structure
Amoako-Gyampah and Acquaah (2008)	The relationship between manufacturing strategy and competitive strategy and their influence on firm performance.	Mediator:
		Quality

Akhtar et al. (2018) and framed as the "capability of a company or an organization to rapidly react to changes and uncertainties" (p. 7294). This definition has been endorsed and used by Wamba (2022), Huang et al. (2021), and Wang et al. (2022), highlighting its broad acceptance in academic discussions. Appendix Q presents different definitions in the literature for agility.

This paper distinguishes between market and operational agilities. Operational agility enables firms to leverage their diverse assets in response to unforeseen industry developments, requiring rapid and proactive business processes to capitalize on opportunities and gain a competitive advantage (Duvivier & Gupta, 2023; Jayawardena et al., 2023). This ability contributes to attaining a

Table 3. Role of AI capability on effectiveness of cost leadership to achieve performance

Study	AI capability	The role of AI capability on effectiveness cost leadership
Ergün and Karabulut (2023); Jang et al. (2020)	Image processing and algorithms, computational capacity, and sophisticated analytical techniques such as deep learning and machine learning.	Mediation
Khan and Alotaibi (2020); Mithas et al. (2020)	AI applications	Antecedent
Chi et al. (2023); Wirtz (2020)	Service robots	Mediation
Jerab and Mabrouk (2023); Vimalkumar et al. (2021)	Automation	Moderation

competitive edge (Cheng et al., 2020; Lee et al., 2015; Mathiassen & Pries-Heje, 2006) and ultimately enhancing organizational performance (Ravichandran, 2018; Sambamurthy et al., 2003; Sharma et al., 2024). Market agility is viewed as an organization's adeptness at utilizing market changes to acquire market intelligence and discern competitive opportunities (Sambamurthy et al., 2003). This crucial competency enhances overall performance (Chatfield & Reddick, 2018; Ravichandran, 2018; Sambamurthy et al., 2003). Additionally, market agility empowers firms to track potential customers in real time, thereby enhancing customer experiences while facilitating real-time monitoring of customer data and personalized information collection (Ravichandran, 2018; Sambamurthy et al., 2003; Wang et al., 2023).

Research Gap

A gap exists in understanding how operational and market agility, as mediating factors, interact with AI application explainability to influence firm performance. Appendix A discusses the relationship between AI capability and firm performance in the current literature. A research opportunity also exists in examining the negative impacts of XAI costs on its performance benefits. Although the literature has explored the association between CLS and firm performance and also looked inro the role of AI capability in the effectiveness of CL to achieve performance, our study addresses a gap by examining the negative moderating role of CL in organizations managing the XAI costs.

Theoretical Backgrounds

Explainability of AI Applications to Generate DC

In today's dynamic business environment, firms must leverage internal and external competencies, focusing on cutting-edge technologies, such as data analytics (Akter et al., 2016), AI (Dwivedi et al., 2021; Wamba et al., 2024), and more recently, XAI. DC theory provides a suitable framework for understanding how organizations develop, integrate, and reconfigure competencies to enhance their performances (Sullivan & Wamba, 2024).

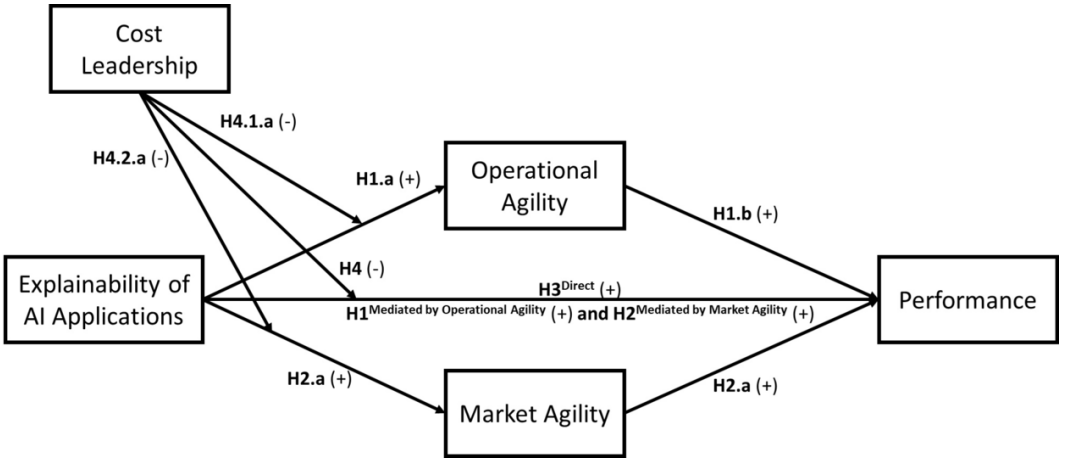
AI applications can provide value beyond their functionality and use. XAI increases an organization's ability to seize opportunities that leverage AI capabilities, such as making data-driven decisions or developing innovative products and services. XAI allows organizations to adapt their internal and external resources and processes in response to business environment changes. Laato et al. (2022) identified the five high-level objectives of explainability of AI applications as understandability, trustworthiness, transparency, controllability, and fairness. Appendix B provides examples of how explainability for AI applications fit the definition of DC.

CT to Support Moderating Role of CL

This study adopts CT to examine the role of CLS in the relationship between XAI, agility, and performance. CT emphasizes the necessity of alignment between key variables for achieving superior performance. Organizations are likely to achieve better performance when they consider contextual factors, thus highlighting the central notion of "good/best fit" (Miller & Matzel, 1988).

CT supports a moderating variable in this study, CL (Putri, 2020). The theory argues that for a firm to achieve competitive advantage and superior performance, its strategic orientation must align with its external context. This alignment, or fit, between the firm and its environment is crucial (Effendi & Kusmantini, 2015). CT advocates for moderating factors to consider relevant environmental factors when studying firm performance. In this study, the theory supports the choice of a CLS. This strategic choice aims to enhance the fit between the firm and its competitive environment, potentially boosting performance (Rasit & Ismail, 2017). Examples in literature have integrated CT into work related to CL (Verma & Aggarwal, 2021). Galbreath et al. (2020) explored the relationship between entrepreneurial orientations and firm performance within the context of low-cost strategies.

Figure 1. Research model



Hypothesis Development

The research model outlined in this study is illustrated in Figure 1.

Mediating Role of Agility

The dynamic capability view (DCV) emphasizes organizational capabilities, particularly agility, in leveraging the benefits of XAI to improve performance. DC refers to "a firm's ability to integrate, build, and reconfigure internal and external competencies to address rapidly changing environments" (Teece et al., 2016, p. 13), helping organizations achieve and sustain a competitive edge (Lin & Kunnathur, 2019; Teece, 2007). DC signifies how a firm is able to (a) sense and shape opportunities and threats, (b) seize market opportunities, and (c) maintain competitiveness through enhancing, aligning, and reconfiguring intangible and tangible resources (Teece et al., 2016). Appendix A provides the definitions in the literature for AI DC. AI provides value beyond its functionality. XAI increases an organization's ability to seize opportunities that leverage AI capabilities. XAI allows organizations to adapt their internal and external resources and processes in response to the business environment changes. Laato et al. (2022) identified the five high-level objectives of explainability of AI applications as understandability, trustworthiness, transparency, controllability, and fairness. Appendix B provides examples of how explainability for AI applications fit the definition of DC. DCV also supports the mediating role of agility in the relationship between explainability of AI applications and performance.

Sambamurthy et al. (2003) highlighted the mediating role of operational agility in performance impacts of IT investments. Werder et al. (2021) suggested that operational agility is a critical factor that mediates the impact of IT investment on firm performance. Literature has also explored market agility as a performance mediator. Previous studies have highlighted customer and market agility as key mediators in the relationship between IT capability-generated customer analytics and firm performance (Roberts & Grover, 2012). They emphasized that agility serves as an intermediary concept that links IT to economic outcomes, providing a unique perspective on how IT contributes to firm value. Studies have also demonstrated that market agility is a strong predictor of new products' success (Hajli et al., 2020) and firm performance (Zhou et al., 2018). Expanding on current literature, we propose mediating roles of operational and market agility in a relation between explainability of AI applications and firms' performances. We propose the following hypotheses:

- H1: Explainability of AI applications within a firm positively impacts performance through the mediating impact of operational agility.
- H1.a: Explainability of AI applications within a firm positively impacts operational agility.
- H1.b: Operational agility positively impacts a firm's performance.
- H2: Explainability of AI applications within a firm positively impacts performance through the mediating impact of market agility.
- H2.a: Explainability of AI applications within a firm positively impacts operational agility.
- H2.b: Market agility positively impacts a firm's performance.
- H3: Explainability of AI applications within a firm positively impacts performance.

Moderating Role of Cost Leadership

This study employs CT as a framework to examine the role of CLS in how explainability of AI applications influence a firm's performance. CL may be effective to improve performance in certain environments where cost efficiency is paramount. CT emphasizes the necessity of alignment between key variables for achieving superior performance. Organizations are likely to achieve better performance when they consider contextual factors, thus highlighting the central notion of "good/best fit" (Miller & Matzel, 1988). CL may hinder agility when the organization must deal with XAI costs. The theory suggests that the degree of contingent factors determines the effectiveness of the explainability and performance relationship, which can be modeled as moderating constructs. Verma and Aggarwal (2021) suggested the moderating role of strategic cost management in business performance. CL's negative moderation could weaken the mediated relationship between explainability, agility, and performance. XAI applications may enhance operational agility, but if the organization is focused on CL, it may not leverage these benefits fully.

The moderating role of CL has been extensively studied in literature. According to Yusuf et al. (2020), the CLS serves as a moderator, enhancing and reinforcing the correlation between supply chain management and financial performance. Alyoubi and Yamin (2021) investigated the moderating role of CL in the relationship between a firm's strategic goals attainment and organizational performance. Al-Rdaydeh et al. (2018) explored how competitive strategy acts as a moderator in the correlation between financial leverage and a firm's performance.

Cost-focused organizations may be less inclined to invest in XAI technologies owing to the perceived high up-front costs and the emphasis on immediate cost savings. This reluctance to adopt XAI could impede an organization's ability to leverage explainability among its AI applications for enhancing operational agility. Having better market agility than competitors is a consistent way to generate value (Salavou & Manolopoulos, 2021). Organizations committed to CLS may exhibit reluctance to allocate resources for XAI. Furthermore, the rigid structure and efficiency-oriented focus inherent in CLS may constrain an organization's adaptability in adjusting market strategies to changing conditions, thereby exacerbating the negative moderating effect of CL on the association between XAI and market agility. Hence, we propose the following hypotheses:

- H4: CL negatively moderates the relationship between explainability of AI applications and performance.
- H4.1: CL negatively moderates the mediated relationship between explainability of AI applications and performance through operational agility.
- H4.2 CL negatively moderates the mediated relationship between explainability of AI applications and performance through market agility.

METHODOLOGY

Data Collection and Measures

We surveyed 157 IT leaders, including C-level executives, in organizations using XAI between November 2023 and March 2024. Participants were selected from LinkedIn and IT leadership forums, ensuring they were over 18 years old and responsible for IT functions. The sample size, determined using G*Power 3 for a 95% confidence level, showed no significant nonresponse bias (Appendix D). We collected data via Qualtrics using a 5-point Likert scale and gathering demographic information. Measurement items, detailed in Appendix C, were adapted from established studies. Control variables included business type (two-digit SIC codes), ownership (public, private, not-for-profit), and organization size (small: fewer than 500 employees, large: 500 employees or more).

To measure AI explainability, we used Arrieta et al.'s (2020) definition that focuses on the quality of explanations based on logical arguments to persuade (Besnard & Hunter, 2008). We adopted a scale for user satisfaction with an explanation based on the study by Hoffman et al. (2023). These attributes of Hoffman et al.'s (2023) scale were combined and formulated into a Likert scale, with the items listed in Appendix C.

On average, respondents reported a tenure of 7.61 years with their current companies. Table 4 shows additional demographic information.

Data Analysis

We used PLS-SEM with Smart PLS 3.3.3 (Ringle et al., 2015) to analyze our models, as recommended by Hair et al. (2017). Before hypothesis testing, we validated the measurement model. The research model (Figure 1) was analyzed using bootstrapping with 10,000 samples, balancing precision and efficiency (Hair et al., 2017). Pilot tests with 10,000 and 15,000 samples showed no significant differences, supporting our choice of 10,000 samples.

To examine moderating and mediating effects, we analyzed CL's role between AI explainability and both operational and market agility and considered agility as a mediator between AI explainability and firm performance. We used multigroup analysis for moderation and the Sobel test for mediation, following Mithas et al. (2011). To enhance reliability, we combined the Sobel test with Preacher and Hayes's (2008) bootstrap method, using the PROCESS macro in IBM SPSS Statistics.

Supplementary information regarding the methods used in this study is provided in Appendix H.

Qualitative Focused Group to Include Global Perspective

To enhance our research to include a more global perspective and to re-examine our quantitative findings through the lens of experts from diverse cultural groups, we conducted a qualitative focus group with 12 C-level executives from diverse regions, including North America, Latin America, Europe, Asia Pacific, the Middle East, and Africa. Appendix I outlines the cultural diversity of our participants based on Gert Hofstede's dimensions of culture. Their extensive experience and varied cultural backgrounds provided valuable insights into the interplay between CL and AI explainability studied in our quantitative survey.

We held an online meeting with these C-level executives to explore how their experiences might have influenced our survey results. Through open-ended questions, we examined the interplay among AI explainability, CL, agility, and organizational performance. The recorded Zoom meetings were analyzed for insights related to our hypotheses.

Table 4. Demographics of participants

Demographic information	Sample (N = 157)
Gender	
Female	48.7%
Male	51.3%
Age	
18–24	0%
25–34	3.2%
35–44	41.6%
45 or older	55.2%
Education	
High school or below	0.7%
Bachelor's degree or college education	52%
Graduate degree	47.3%
Company size	
Small-to-medium (fewer than 500 employees)	48.1%
Large (500 or more employees)	51.9%
Main line of business (categories using two-digit SIC codes)	
Agriculture, forestry, and fishing (01–09)	2.5%
Mining (two-digit SIC: 10–14)	0%
Construction (two-digit SIC: 15–19)	1.3%
Manufacturing (two-digit SIC: 20–39)	21.7%
Transportation and public utilities (two-digit SIC: 40–49)	17.6%
Wholesale trade (two-digit SIC: 50 and 51)	9.1%
Finance, insurance, and real estate (two-digit SIC: 52–67)	27.4%
Services (two-digit SIC: 68–78)	16.1%
Public administration (two-digit SIC: 79–85)	4.3%
Nonclassified establishments (two-digit SIC: 86)	0%

RESULTS

Assessment of Measurement Model

We assessed the convergent and discriminant validity to evaluate the measurement model. Although we used established instruments from existing literature in our questionnaire, we consider our instrument to be a unique measure that needs to be evaluated for unidimensionality. Unidimensionality examines whether all items in a scale contribute to measuring the intended construct (Gerbing & Anderson, 1988). Because PLS does not directly assess unidimensionality, we used SPSS to perform a scree test (D'agostino & Russell, 2014) to determine the number of latent constructs in the dataset. Following this test, we used SmartPLS to examine the questionnaire items for loading, weight, validity, and reliability. The factor analysis results for the constructs are shown in Table 5.

We assessed whether item loadings were significant for inclusion in a specific construct by examining construct-to-item loadings for reflective constructs. Following the study by Hair et al.

Table 5. Convergent and discriminant validity of measurement model

Construct	CR	AVE	Factor correlation				
			1	2	3	4	5
1. Explainability of AI applications	0.87	0.71	**0.86**				
2. CL	0.95	0.73	0.74	**0.87**			
3. Operational agility	0.92	0.75	0.63	0.69	**0.85**		
4. Market agility	0.91	0.53	0.71	0.63	0.68	**0.91**	
5. Performance	0.94	0.79	0.51	0.49	0.71	0.88	**0.90**

Note. CR = Composite reliability; AVE = Average variance extracted

(2011), only items with loadings exceeding 0.70 were retained, consistent with the scree test results. Subsequently, we calculated the composite reliability (Raykov, 1998) to assess the internal consistency of each reflective construct. Reliability scores above 0.70 were acceptable in early research stages, whereas a threshold of 0.80 was recommended for well-established constructs (Nunnally, 1978). Table 6 provides an overview of construct validity, reliability, and correlation matrices, with each construct showing reliability exceeding 0.80. Additionally, in Table 6, the square root of the AVE exceeded off-diagonal correlations, indicating good construct validity. Following the study by Fornell and Larcker (1981), we assessed the square root of the AVE with the correlation between each pair of constructs to evaluate construct validity. Table 6 shows that, for each construct, the square root of the AVE was greater than the construct's off-diagonal correlations. Furthermore, we conducted several variance inflation factor tests, and all these tests returned a maximum value of 2.1, well below the threshold of concern (10) for reflective constructs (MacKenzie et al., 2011). These results confirm the good discriminant validity of all scales.

Structural Model

We examined the path coefficients (β) to depict the strengths of the relationships posited in our hypotheses. We used R^2 to measure the proportion of variance in the dependent variable explained by each independent variable. We used both path coefficients and R^2 to comprehensively assess the model's performance. Hypothesis H1.a, suggesting a positive relationship between explainability of AI applications and operational agility, was supported ($\beta = 0.33$, $p < .001$). H1.b, which posits a positive association between operational agility and firm performance, was substantiated ($\beta = 0.28$, $p < .05$). H2.a, proposing a positive link between explainability of AI applications and market agility, was also confirmed ($\beta = 0.31$, $p < .001$). H2.b, suggesting that market agility positively influences performance, was supported ($\beta = 0.63$, $p < .001$). However, H3, stating that explainability of AI applications has a positive direct association with firm performance, was not supported by our analysis ($\beta = 0.06$, $p > 0.05$).

In summary, the two types of agility account for 74.11% of the variance in operational agility and 73.89% of the variance in market agility. These two forms of agility collectively explain 82.34% of the variance in firm performance.

We examined the mediation effects of operational agility and market agility in the relationship between explainability of AI applications and performance using the PROCESS macro. The results indicate significant mediation for both operational agility (mediated effect = 0.29, $p < 0.001$, lower CI: 0.08, Upper CI: 0.37) and market agility (mediated effect = 0.38, $p < 0.01$, lower CI: 0.31, upper CI: 0.6). Indirect effects were deemed statistically significant if the 95% confidence interval did not include zero, indicating the presence of mediating effects. Although the direct effect of explainability of AI applications on firm performance (H3) was insignificant, Table 7 shows that the relationship

Table 6. Results of hypothesis testing of direct effects

Hypothesis	β value
H1.a: Explainability of AI applications → operational agility	0.33**
H1.b: Operational agility → performance	0.28*
H2.a: Explainability of AI applications → market agility	0.31***
H2.b: Market agility → performance	0.63***
$H3^{Direct}$: Explainability of AI applications → performance	$0.06^{N/S}$
Variance Explained (R^2)	
Operational agility	71.51%
Market agility	70.19%
Performance	74.14%

*** p < 0.001, ** p < 0.01, * p < 0.05, N/S: The relationship is not significant.

Table 7. Mediation effects

Hypothesis	Mediated effect	Lower CI	Upper CI	PME
H1: Explainability of AI applications →Operational agility → Performance	0.29***	0.08	0.37	0.45
H2: Explainability of AI applications →market agility → performance	0.38**	0.31	0.64	0.55

*** p < 0.001, ** p < 0.01, * p < 0.05

between explainability of AI applications and performance is mediated by both operational and market agilities.

The proportion of mediated effect (PME) values for operational agility and market agility provide important insights into the relative contribution of these mediating variables in the relationship between explainability of AI applications and performance. A PME value of 0.45 for operational agility indicates that 45% of the total effect of explainability of AI applications on performance is mediated by operational agility. This finding suggests that nearly half of the impact of explainability of AI applications on performance can be attributed to its influence on operational agility, highlighting the significant role that operational agility plays in translating implementation of explainability among AI applications into improved performance outcomes. Similarly, the PME value of 0.55 for market agility indicates that 55% of the total effect of explainability of AI applications on performance is mediated by market agility. This finding implies that more than half of the impact of explainability of AI applications on performance can be attributed to its influence on market agility, underscoring the importance of considering market agility in conjunction with implementation of explainability among AI applications.

To assess the moderating effect of CL on the relationship between explainability of AI applications and various outcomes, we conducted moderated hypotheses tests using the same PROCESS macro procedure. The results revealed a significant negative moderation effect of CL on the direct association between explainability of AI applications and performance (H4: Explainability of AI applications × CL → performance), indicating that higher levels of CL significantly decrease performance in the presence of increased explainability of AI applications. This moderation is evidenced by the significant interaction term (moderated effect = -0.25, $p < 0.01$, lower CI: -0.35, upper CI: -0.15), suggesting that the impact of explainability of AI applications on performance varies based on the level of CL. In organizations that do not prioritize CL, the direct association between explainability of

Table 8. Moderation effects

Hypothesis	Moderated effect	Lower CI	Upper CI	ΔR^2
H4: Explainability of AI applications × CL → performance	-0.25**	-0.35	-0.15	0.12
H4.1.a: Explainability of AI applications × CL → operational agility	$-0.18^{N/S}$	-0.28	-0.08	0.08
H4.2.a: Explainability of AI applications × CL → operational agility	$-0.21^{N/S}$	-0.31	-0.11	0.09

*** p < 0.001, ** p < 0.01, * p < 0.05, N/S: The relationship is not significant.

AI applications and performance is insignificant. However, in organizations with a stronger focus on CL, increased explainability of AI applications is significantly associated with decreased performance.

CL moderation effect was also observed in the relationships between explainability of AI applications and operational agility (H4.1.a: Explainability of AI applications × CL → operational agility) and explainability of AI applications and market agility (H4.2.a: Explainability of AI applications × CL → market agility). In both cases, higher levels of CL not only weakened these relationships but also rendered them insignificant. This finding indicates that the influence of explainability of AI applications on operational agility and market agility diminishes in organizations with higher levels of CL.

The results in Table 8 show that CL moderating effect is significant (lower and upper CIs do not include zero) and negative for all hypotheses. For the relationship between explainability of AI applications and performance, the interaction term (moderated effect = -0.25) is significant ($p < 0.01$), indicating a negative impact of CL on this relationship. The model for performance explains a substantial amount of variance, with CL explaining R-squared change (ΔR^2) of 0.12 that demonstrates an additional 12% of the variance. We observed similar patterns for the relationships between explainability of AI applications and operational agility (moderated effect = -0.18, $p > 0.05$, $\Delta R^2 = 0.08$) and market agility ($\beta = -0.21$, $p > 0.05$, $\Delta R^2 = 0.09$).

In Table 9, the moderated mediated hypotheses also demonstrate the significant and negative moderation effects of CL. The indices of moderated mediation (IMM) for both hypotheses indicate a weakening and insignificant effect of explainability of AI applications on performance through the mediating variables (operational agility and market agility) in organizations with higher levels of CL. Specifically, for hypothesis H4.1, the moderated mediation effect is -0.17 ($p > 0.05$), with a 95% confidence interval (CI) ranging from -0.27 to -0.07, which does not include zero. Similarly, for hypothesis H4.2, the moderated mediation effect is -0.19 ($p > 0.05$), with a 95% CI ranging from -0.29 to -0.09. These results suggest the significant moderated mediation effect as lower and upper CLs do not include zero. The findings also illustrate that while explainability of AI applications may enhance performance through operational agility and market agility in organizations with lower levels of CL, this relationship becomes insignificant in organizations where CL is prioritized. These findings highlight the nuanced role of CL in moderating the mediated impact of explainability of AI applications on organizational outcomes through operational and market agility, emphasizing the need for organizations to consider the extent of their focus on CL when implementing XAI.

Our analysis of the control variables reveals that none of the control variables has a significant effect.

Discussion Qualitative Results From Focus Group

The global perspectives shared in our focus group discussions align with the quantitative survey findings on the role of XAI and CL in agility and performance. Appendix K maps the key terms in our qualitative data to the constructs in the quantitative study. Appendix J provides sample responses.

Table 9. Moderated mediated effects

Hypothesis	Moderated mediation effect	Lower CI	Upper CI	IMM
H4.1: Explainability of AI Applications × CL → Operational Agility → Performance	-0.17$^{N/S}$	-0.27	-0.07	-0.15
H4.2: Explainability of AI Applications × CL → Market Agility → Performance	-0.19$^{N/S}$	-0.29	-0.09	-0.18

*** p < 0.001, ** p < 0.01, * p < 0.05, N/S: The relationship is not significant.

Appendix O provides additional results from our focus group about the cost of XAI and ethical considerations in XAI.

Mediating Role of Operational and Market Agility

The examples from the United States, Germany, and India illustrate how XAI enhances operational agility by enabling organizations to adapt swiftly to environmental disruptions, regulatory changes, and infrastructure challenges. These practical applications align with research highlighting XAI's role in optimizing supply chain management, production scheduling, and resource allocation processes.

In Japan, Saudi Arabia, and Australia, the focus group discussions underscored how XAI supports market agility by providing clear insights into consumer behavior, enabling proactive adjustments in marketing strategies and investment decisions. These insights corroborate survey findings on XAI's impact on market responsiveness and competitive positioning in dynamic business environments

Moderating Role of Cost Leadership

Conversely, the challenges posed by CL strategies in Brazil, China, and Nigeria emphasize the limitations of XAI when organizational priorities prioritize cost efficiency over innovation and agility. These examples provide practical context to the survey's findings on the negative moderating effect of CL on XAI's potential to enhance overall performance and competitiveness.

DISCUSSION

This research examined how XAI impacts company performance. Our study, with data from a quantitative survey of 157 executive IT managers, as well as a qualitative focus group of 12 global executives, supports that XAI enhances performance through the mediation of operational (H1) and market (H2) agility.

Although existing research (Behera et al., 2023) and our study acknowledge the performance benefits of XAI, there is a gap in understanding the costs associated with XAI and its impact on performance. This study addresses the gap by examining the moderating role of CL in managing XAI costs and its potential negative impact on performance. Our findings indicate that although XAI explainability does not directly relate to performance, its benefits are realized through operational and market agility. However, in organizations prioritizing CL, the positive relationship between explainability and performance becomes negative, and the mediating effects of agility on performance become insignificant. Therefore, in CL-focused organizations, XAI may hinder rather than enhance performance.

Theoretical Implications in Realm of Global Information Management

This study makes three contributions to literature of global information management. This study contributes by highlighting the importance of XAI for improving the market and operational agility to achieve performance in global information management. It aligns with previous work on

competitive advantage of XAI in global information systems (Behera et al., 2023). The study also adds to the literature on AI's performance outcomes through a mediating role of agility (Akter et al., 2016; Dubey et al., 2003; Wamba, 2022; Wang et al., 2022), portraying it as a dynamic capability enabled by XAI. Unlike prior studies focusing on existing AI capabilities, this research investigates how organizations can manage evolving XAI applications to adapt to dynamic markets, which is the necessity of today's global information management environment.

Second, the study examines how market and operational agility's mediating influence, as two significant characteristics of global information management, remains significant regardless of infrastructure requirements that has been an issue in global context when managing information infrastructure. Previous research assumed AI infrastructure was managed internally, requiring substantial capital investments, which in the context of global information management is unrealistic. This study challenges the assumption when integrating XAI into a firm's work systems, proposing a model where explainability of AI impacts firm performance through agility, independent of IT infrastructure.

Third, this study explores the moderating impact of the CLS on agility-mediated performance of XAI in information management practices of companies with a global context. It investigates whether this strategy, as a characteristic of global information management context, negatively affects agility at the market and operational levels. Prior research (Amoako-Gyampah & Acquaah, 2008; Hilman & Kaliappen, 2014; Kaliappen & Hilman, 2013; Kharub et al., 2022; Nandakumar et al., 2010; Voola & O'Cass, 2010) extensively explored the correlation between implementing a CLS and its positive effects on firm performance. In particular, the literature highlights the established positive moderating effects of CL on overall organizational performance (Al-Rdaydeh et al., 2018; Alyoubi & Yamin, 2021; Yusuf et al., 2020). However, this study demonstrates disagreement with these findings by looking into the less explored negative moderating role of CL in organizations dealing with the costs associated with XAI. Our study posits that within organizations that prioritize CL as a competitive strategy, the explainability of AI applications has an adverse effect on their performance. This study underscores the importance of considering the moderating effects of CL, in the context of global information management, for AI implementation, highlighting the need for a nuanced understanding of the relationship between explainability of AI applications, CL, and organizational performance.

Practical Implications in Realm of Global Information Management

From a practical perspective, this article has two implications for organizations seeking to enhance their performance when managing information management through XAI applications in a global context. First, improving AI explainability enhances operational agility by enabling quicker decision-making and adaptive responses to challenges that are significantly important in a global context of information management. For instance, XAI provides visibility into inventory, demand forecasts, and supplier metrics that are important parameters in companies in global operation. In manufacturing, XAI optimizes production schedules using real-time data and predictive analytics, aligning outputs with market demands and resource availability to reduce lead times and improve product quality. Similarly, XAI enhances market agility by analyzing consumer trends, facilitating rapid adjustments to product offerings. In financial services, XAI supports risk management and investment decisions by clarifying risk assessments and portfolio recommendations, enabling firms to adapt swiftly to market changes and optimize returns amid regulatory shifts and economic fluctuations. In conclusion, prioritizing AI explainability empowers organizations to enhance operational and market agility, driving significant improvements in overall performance outcomes. Organizations can leverage XAI to align operational decisions with business objectives and market dynamics to improve performance.

Second, the study shows that organizations focused on CL must carefully navigate the balance between cost efficiency and XAI particularly when information management infrastructure is globally being considered with differences among countries in their financial stability and

economical capabilities. These organizations typically prioritize cost efficiency in their global markets for information management, but they may face challenges in maintaining agility during XAI implementation. Therefore, conducting a thorough cost-benefit analysis to optimize XAI investments effectively is crucial for these organizations. By doing so, they can align these investments with strategic objectives and operational capabilities, thereby mitigating the potential risk of sacrificing agility for immediate cost savings. This strategic approach ensures that organizations sustain their competitiveness in rapidly evolving markets while harnessing the transformative potential of XAI to enhance operational efficiency and strategic agility. However, balancing these efficiency gains with initial investment outlays is crucial to maintain CL without compromising agility in responding to operational and market shifts.

Limitations and Future Work

We identified eight avenues for future improvement. First, our reliance on self-reported data from IT executives limits causal inference that was addressed by assessment of measurement model. One construct that could be particularly affected by self-reported data is firm performance. To mitigate this, in Appendix E we cross-validated firm performance using archival data. Social desirability bias as another limitation of self-reported data is addressed in Appendix F. Second, we did not explore how XAI explainability impacts agility. Therefore, follow-up studies could explore the underlying mechanisms of this relationship. Third, we focused on operational and market agility, but there are broader dimensions of agility to consider. Fourth, future research could explore other potential mediators beyond agility, such as IT governance or leadership attributes. Fifth, our study opted for a self-selected sample primarily owing to practical constraints in accessing a specific population of interest. Participants were selected from organizations that had actively used XAI. Although we recognize that a randomized sample would have helped mitigate potential biases, the nature of our study and the focus on XAI make the self-selected sample a pragmatic and informative choice. Sixth, our study lacks a longitudinal analysis of XAI's impacts; such an analysis could offer insights into its evolving influence on firm performance. Seventh, our dataset provides valuable insights into the relationships under the study, but it is limited in its coverage of diverse industries. Our sample primarily includes data from industries that we have indicated in Table 5. This limitation suggests that the findings may not be fully generalizable to industries outside this scope. To enhance the generalizability of the findings, future research should consider using datasets that include a wider variety of industries. Such an approach would help determine whether the observed relationships hold across different sectors. Eighth, our study primarily examines the cost implications of AI explainability within the context of CL. To enrich the analysis of AI explainability, future research should consider investigating its implications under alternative strategic frameworks, such as differentiation and focus strategies.

CONCLUSION

Many global companies are integrating XAI to gain competitive advantages in their information management practices. This study investigates how explainability of AI applications within a firm can enhance its performance through agility. Our findings demonstrate the negative impact of the CLS on both operational and market agility in the context of global information management. Our study suggests that organizations pursuing CLS should consider the potential trade-offs with operational and market agility when seeking to improve their performance through explainability of AI. In summary, our study offers valuable insights into leveraging XAI to enhance organizational operations, highlighting the importance of addressing cost concerns with XAI when operating for global information management.

CONFLICTS OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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